



Fig. 12: Rear side of the cabinet

jack and power cable to a USB type-A cable for external connections. Therefore, solder all the cables to respective connectors, as shown in Fig. 7 and wrap them with cello tape, as shown in Fig. 8.

Connecting the cables to MP3 player. The 3-wire cable is used for stereo audio connections, the 2-wire cable is for power supply connection, and the single-strand cable is for antenna. The inside view of the author's prototype with various cables is shown in Fig. 9.

Note that there is a 4-pin top entry header on the PCB of the MP3 module. But the module comes with a 3-pin wire for stereo audio connection. You should connect it to first three pins of



Fig. 13: Multiport audio jack



Fig. 14: Complete assembly of Bluetooth MP3 player and FM receiver

the 4-pin header from right towards the 2-pin red header connector for power supply. The last pin is left unused, as shown in Fig. 9.

Before fixing the module to the cabinet, ensure everything is working properly. When you change the mode in IR remote, it should change to five modes one by one: bt (Bluetooth), USB, FM, SD, and AUX. Testing Bluetooth function in the MP3 player module is shown in Fig. 10.

Fixing MP3 player in the cabinet. Fixing the MP3 player module in the cabinet is a no-brainer. From the front side, insert the module into the 8cm × 1.7cm slot cut out in the cabinet. Fix it using nuts and bolts. Connect the cables inside the cardboard cabinet, as shown in Fig. 11.

All the cables can be extended to different points from the holes provided on the rear side of the cabinet, as shown in Fig. 12. You can connect to audio source or amplifier through a 3.5mm multiport audio jack, as shown in Fig. 13.

The complete assembly of the Bluetooth MP3 player with FM receiver is shown in Fig. 14. The MP3 player module can be powered by connecting it to a 5V DC USB power supply. Its output can be connected to an audio amplifier or amplispeaker to start listening to music. A small desktop stereo speaker system was used to test the prototype.

The MP3 player module has USB and MicroSD ports to play MP3 files. Audio from mobile phone or laptop can be listened to by connecting through Bluetooth. FM radio can also be listened to.

The MP3 player can be controlled using IR remote or control buttons provided on the front panel. It also has an AUX (auxiliary) socket, which can be plugged to a 3.5mm standard jack headphone connection. **EFY**

Sani Theo, technical editor at EFY, is an electronics enthusiast and circuit designer